Virtualisation Enhancements in Fedora 12

Release 12, the latest Fedora distribution that came out in November 2009, offers users a lot of new virtualisation features. The Dirty Dozen, as this set of virtualisation features is called, are targeted at improving security, better performance, more stability, a better platform for developers as well as some usability enhancements.

In this article, we'll look at the components that make up the virtualisation stack to run KVM-based virtual machines on Fedora. At the lowest level of the virtualisation stack, we have the Linux kernel that functions as the hypervisor. The QEMU project presents the device model, or the virtual computer to guest operating systems. The libvirt library manages various hypervisor and emulator backends and presents a consistent and stable API allowing user interface apps to interact with VMs. Finally, the virt-manager UI is used to manage the virtual machines.

This stack, comprising the Linux kernel, QEMU, libvirt and virt-manager is the recommended way of running virtual machines on top of a Fedora installation.

According to Fedora policy, all work happens in upstream projects. Fedora then automatically inherits all the goodies that come along with software updates. To enlist some of the new features that got introduced in Fedora 12 in relation to the Fedora 11 release, let's look at how these projects have evolved. The following table shows the version of the package first included in Fedora 11 and the version that's included in Fedora 12:

	Version in Fedora 11	Version in Fedora 12
Linux kernel	2.6.29	2.6.31
QEMU	0.10.4	0.11.0
libvirt	0.6.2	0.7.1
virt-manager	0.7.0	0.8.0

Note that Fedora 11 currently has kernel version 2.6.30 as part of the updates, and the other packages are updated to incorporate bug fixes from upstream releases, as well as user bug reports.

Let's now look at each of these components to identify the key changes that lead to a better experience for users.

The Linux Kernel: The key component for virtualisation in the Linux kernel is the KVM code that enables the Linux kernel to function as a hypervisor. The KVM changes include:

- Performance enhancements in the guest interrupt injection logic